Total no of page-03

Roll No----------

End-Semester Examination (Old Scheme Prior to 2015), July' 2021

B. Tech. (Eighth Semester)

HU-413.1- Econometrics (University Elective)

Marks - 70 Marks
Time- 3.00 Hrs

Attempt any five questions.
Assume missing value, if any.

1. a	What do you mean by Econometrics? Discuss its relevance for engineers.	7
1. b.	Ten industries of the State have been ranked according to profit earned in 2015-16 and the working capital for that year. Calculate the rank correlation coefficient.	7

Industry	A	B	C	D	E	F	G	H	I	J
Rank (Profit)	1	2	3	4	5	6	7	8	9	10
Rank (working Capital)	10	1	6	5	8	2	4	3	7	9

2. a	Discuss significance of Assumptions of Multiple Linear Regression	7
2. b	What do you mean by stochastic variable? Discuss significance of stochastic variable.	7
3. a	Suppose you collect data from a survey on wages, education, experience, and gender. In addition, you ask for information about cigarette smoking. The original question is: "On how many separate occasions last month did you smoke cigarettes? Write an equation that would allow you to estimate the effects of cigarette smoking on wage, while controlling for other factors. You should be able to make statements such as, "Cigarette smoking five more times per month is estimated to change wage by x%".	6
3. b	Write a model that would allow you to test whether cigarette smoking has different effects on wages of man and women. How would you test that there are no differences in the effects of cigarettes smoking for men and women?	4
3. c	Suppose you think it is better to measure cigarette smoking by putting people into one of four categories: non user, light user (1-3 cigarettes per day), moderate user (3-6 cigarettes per day), heavy smoker (more than 6 cigarettes	4

per day). Now, write a model that allows you to estimate the effects of cigarette smoking on wage.

4. a The table below shows the average annual percentage rates of growth of employment, E and real GDP, g, for 25 OECD countries for the period 20XX-20YY. The regression output shows the result of regressing E on g. Provide an interpretation of the coefficients. 7

Average annual percentage rates of growth of employment and real GDP, 20XX-20YY:

Country	Employ.	GDP	Country	Employ.	GDP
Australia	1.68	3.04	Korea	2.57	7.73
Austria	0.65	2.55	Luxembourg	3.02	5.64
Belgium	0.34	2.16	Netherlands	1.88	2.86
Canada	1.17	2.03	New Zealand	0.91	2.01
Denmark	0.02	2.02	Norway	0.36	2.98
Finland	-1.06	1.78	Portugal	0.33	2.79
France	0.28	2.08	Spain	0.89	2.60
Germany	0.08	2.71	Sweden	-0.94	1.17
Greece	0.87	2.08	Switzerland	0.79	1.15
Iceland	-0.13	1.54	Turkey	2.02	4.18
Ireland	2.16	6.40	United Kingdom	0.66	1.97
Italy	-0.30	1.68	United States	1.53	2.46
Japan	1.06	2.81	-	-	-

4. b How we can use the t test for determining the statistical significance of the slope of the regression line? 7

5. a Explain the concept of multiple regressions and explain its importance in managerial decision making. 7

5. b How White's test is an improvement over Goldfeld-Quandt test to discuss Heteroscedasticity? Discuss with suitable example. 7

6 Comment on the following statement: 14

6. a The management of M/s XYZ co. LTD stated in a special bulletin"……: The average salary paid to the workers in 2015 is nearly double that of 2005. So, our workers are enjoying a standard of living 100% higher in 2015 compared to 2005……".

6. The average price of certain commodities in 2011 was Rs. 40 and the average

b	price of same commodities in 2012 was Rs. 80. Therefore, the increase in 2012 on the basis 2011 was100%. So, the decrease should have been 100% in 2011, using 2012 as base.
6. c	From the following prices of different years, which year should be considered as reasonable base and why?

Year	2009	2010	2011	2012	2013	2014	2015
Prices	50	100	110	115	250	120	125

6. d	The government wants to increase the Dearness Allowance in the year 2017 to the employees for which a suggestion has been given that the year 1950 should be taken as base year for construction of Index Number.
6. e	The Directorate of a company want to pay the Dearness Allowance in the year 2017 to their employees, whom they have already paid Dearness Allowance in 2014. For the construction of Index Number, which year should be taken as base year and why?
6.f	Will it be reasonable to include the prices of Television sets and Refrigerators in the construction of Cost of Living Index Number for workers? Give reasons in support of your answer.
6. g	During a certain period, the cost of Living Index Number goes up from 110 and 200 and the salary of a worker is also raised from Rs. 3025 to 3500. Does the worker really gain, and if so, by how much in real terms?
7. a	Show how measures of dispersion help in explaining that though frequency distributions may have the same values of their average, they may differ in their respective formation. In what respect are measures of dispersion of use i statistics?
7. b	1. Discuss following
i.	where mode helps, median is useless
ii,	Where GM help, AM is uselss
iii.	Can two Median value in a distribution.
iv.	Where weight should be given to the value?